

Question 1 – ABC Fund vs Benchmark and Peers

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Question 1 overview

- Compare **ABC Fund** against
 - the **FTSE/JSE Capped SWIX** benchmark
 - the **average active manager** (peers)
- Main questions
 - How does ABC's **beta** compare to peers over time?
 - How much of ABC's returns are explained by the benchmark (single factor model, R-squared)?
 - How does ABC's **performance** compare (cumulative and rolling returns)?
 - How does ABC's **risk** (volatility) compare to peers and the benchmark?
- Since we have many managers, we use the **peer average** rather than every individual peer.

We will use the **analysis data** we construct below for most graphs and tables, and then extend this in later questions.

Libraries and helper functions

Raw data

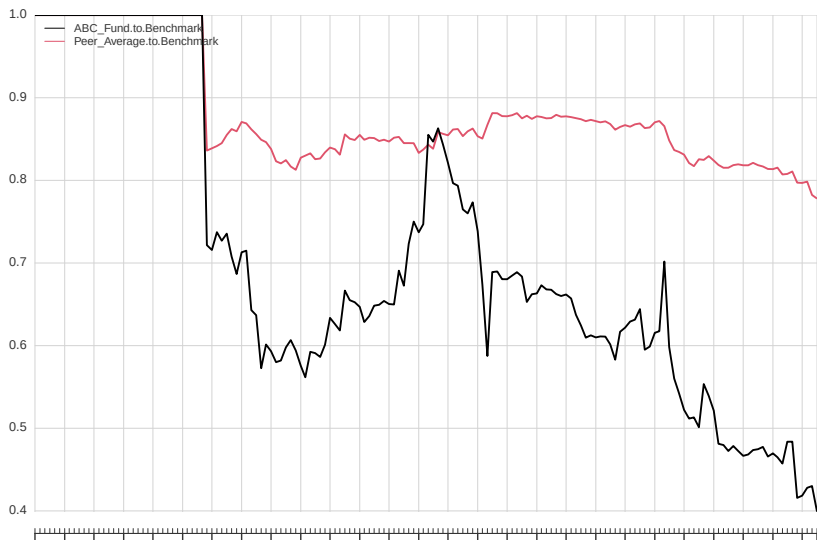
Common sample and analysis data

```
## Rows: 160
## Columns: 6
## $ date                <date> 2012-07-31, 2012-08-31, 2012-
## $ ABC_Returns          <dbl> 0.0349878, 0.0303066, 0.006714
## $ Benchmark_Returns    <dbl> 0.03192534, 0.02709145, 0.0119
## $ Peers_Avg_Returns    <dbl> 0.021872081, 0.016157091, 0.01
## $ Peers_Median_Returns <dbl> 0.02155600, 0.01871670, 0.0159
## $ Peers_Count          <int> 88, 87, 88, 88, 87, 88, 90, 90
```

Rolling 3 year beta vs benchmark

Rolling 3-year beta vs benchmark: ABC vs peer average

2012-07-31 / 2025-10-31



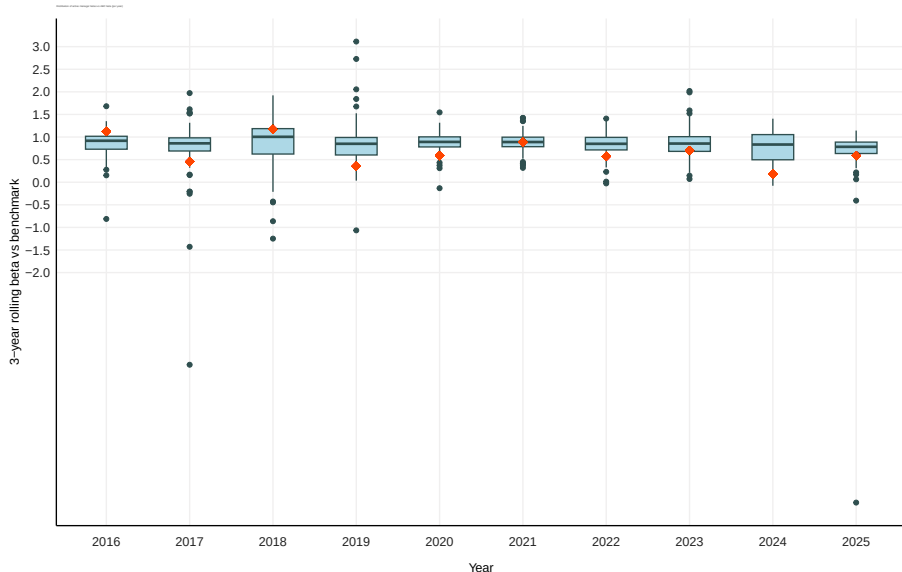
Slide – interpretation of rolling beta

- The chart shows **time varying betas** for:
 - ABC Fund
 - Peer average
- We can comment on:
 - Periods when **ABC takes more or less market risk** than peers
 - Whether ABC's beta is **consistently above / below** the peer average
 - How stable the peer average beta is compared to ABC

You can add a few bullet points here based on what your actual plot shows.

3 year rolling beta distribution per year

3-year rolling beta: Active managers vs ABC Fund



Note:

The orange diamond shows the ABC Fund's 3-year rolling beta

Slide – interpretation of beta boxplots

- Each boxplot summarises the **distribution of peer betas** for a given year.
- The **orange diamond** shows the **ABC Fund beta** for that year.
- Comment on:
 - Does ABC usually sit **inside** the peer distribution, or is it often an **outlier**?
 - Are there years where ABC takes **materially more or less market risk** than most managers?

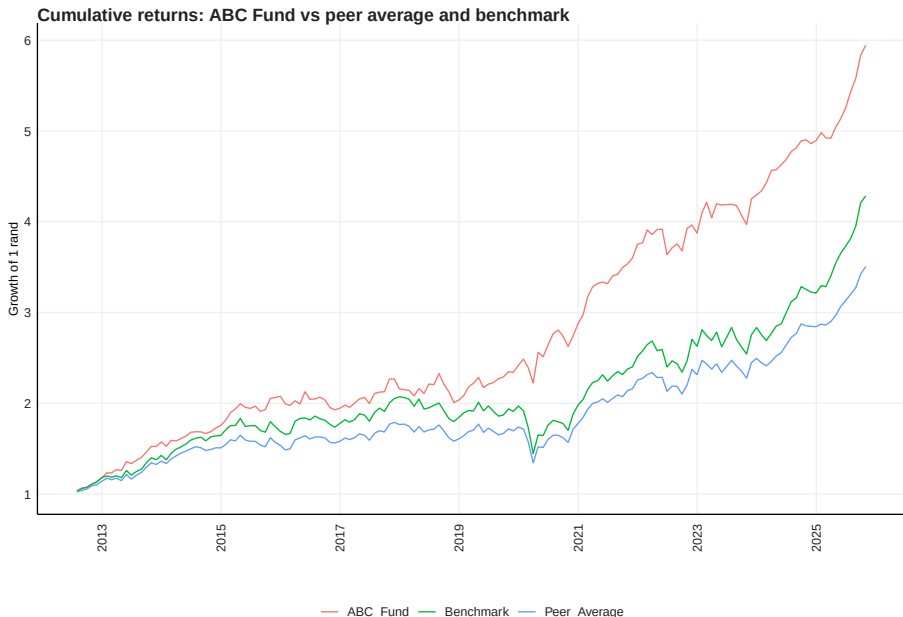
Summary statistics and single factor model

##	Benchmark	ABC_Fund	Peer_Average
## Observations	160.0000	160.0000	160.0000
## NAs	0.0000	0.0000	0.0000
## Minimum	-0.1669	-0.0715	-0.1486
## Quartile 1	-0.0150	-0.0040	-0.0130
## Median	0.0141	0.0123	0.0127
## Arithmetic Mean	0.0099	0.0117	0.0084
## Geometric Mean	0.0091	0.0112	0.0079
## Quartile 3	0.0321	0.0290	0.0268
## Maximum	0.1418	0.1509	0.1303
## SE Mean	0.0030	0.0024	0.0026
## LCL Mean (0.95)	0.0039	0.0069	0.0033
## UCL Mean (0.95)	0.0158	0.0165	0.0135
## Variance	0.0014	0.0009	0.0011
## Stdev	0.0379	0.0308	0.0329
## Skewness	-0.4396	0.2528	-0.4085
## Kurtosis	2.8386	2.0787	3.3731

Slide – interpretation of factor model

- Use **table.SFM** to comment on:
 - **Alpha** of ABC vs benchmark
 - **Beta** estimates and their standard errors
 - **R-squared** to show how much of ABC's return is explained by the benchmark
- Compare ABC to **peer average**:
 - Is ABC's **alpha higher or lower** than peers?
 - Does ABC's **R-squared** suggest it behaves more like a benchmark hugger or takes more idiosyncratic bets?

Cumulative returns – ABC vs peers vs benchmark



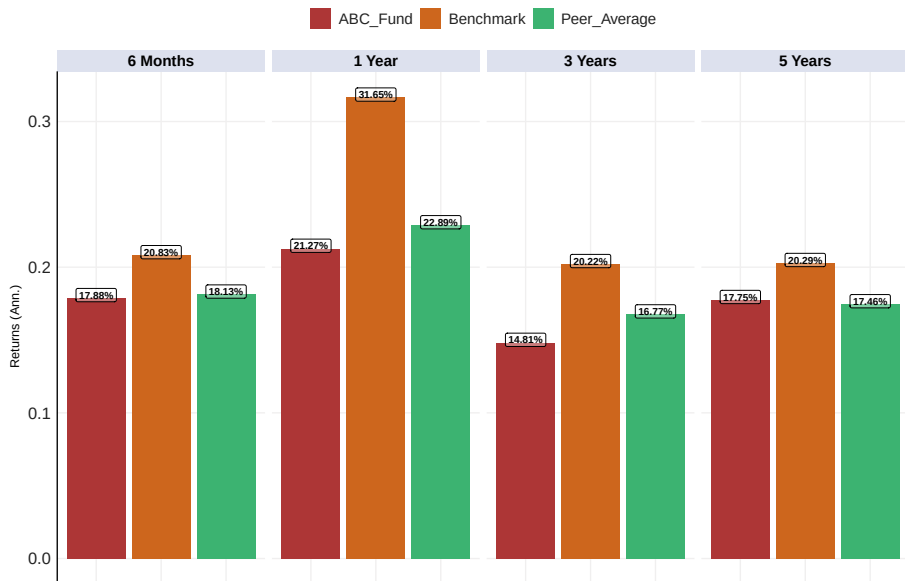
Slide – interpretation of cumulative performance

- Compare the **terminal value** of 1 rand invested in:
 - ABC Fund
 - Peer average
 - Benchmark
- Comment on:
 - Does ABC **outperform** or **underperform** over the full sample?
 - Are there long periods where ABC **tracks peers** or **diverges** strongly?

Rolling 3 year annualised returns

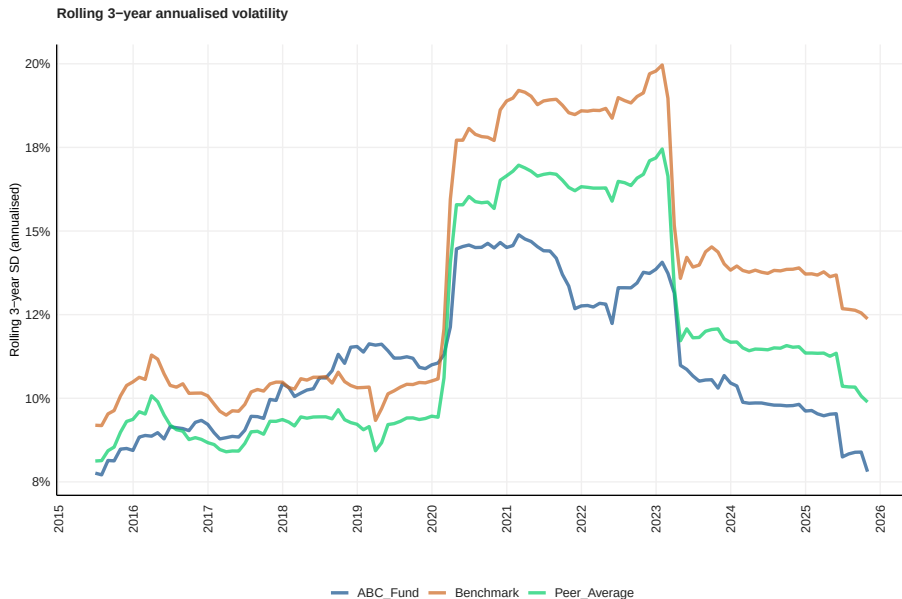


Multi horizon annualised return bar chart



Note:

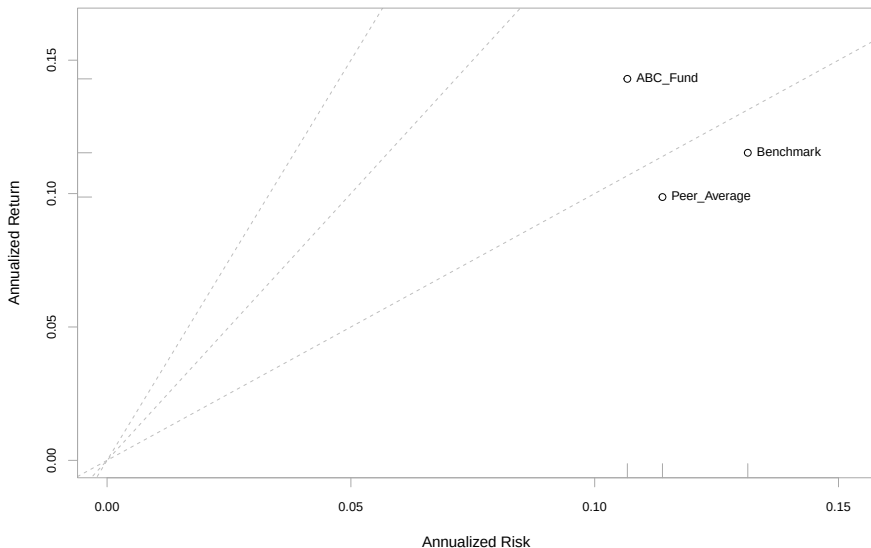
Rolling 3 year volatility



Note:

Risk return scatter and statistics table

Risk–return scatter: ABC Fund vs peer average vs benchmark



Closing slide – summary for Question 1

- ABC Fund vs benchmark and peers in terms of:
 - **Beta and risk** (time varying and cross section)
 - **Performance** (cumulative, rolling and at different horizons)
 - **Factor exposure** and **R-squared** relative to the market